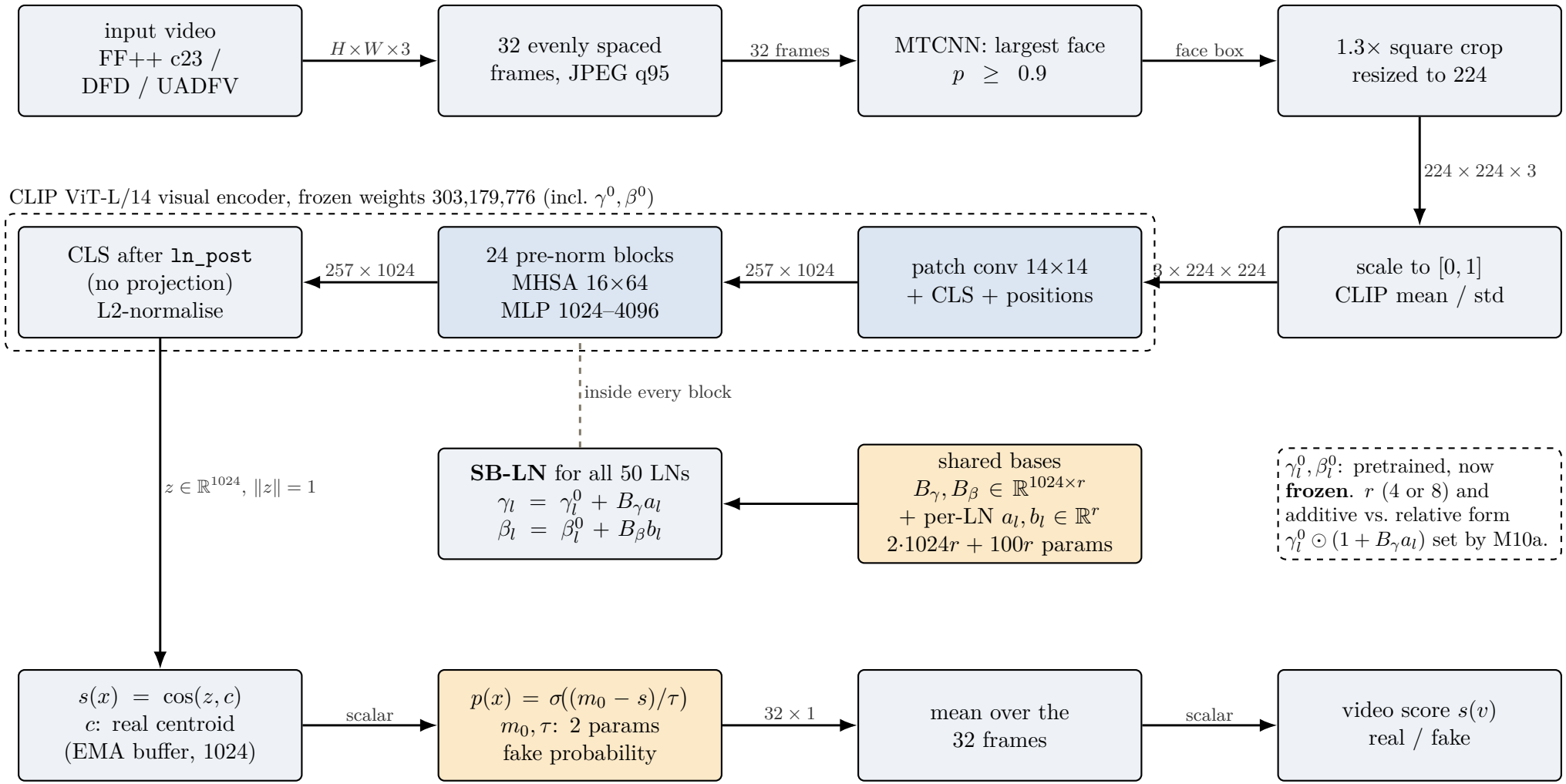
Proposed Architecture — Inference Pipeline (SB-LN + Real-Anchored Cosine Head)

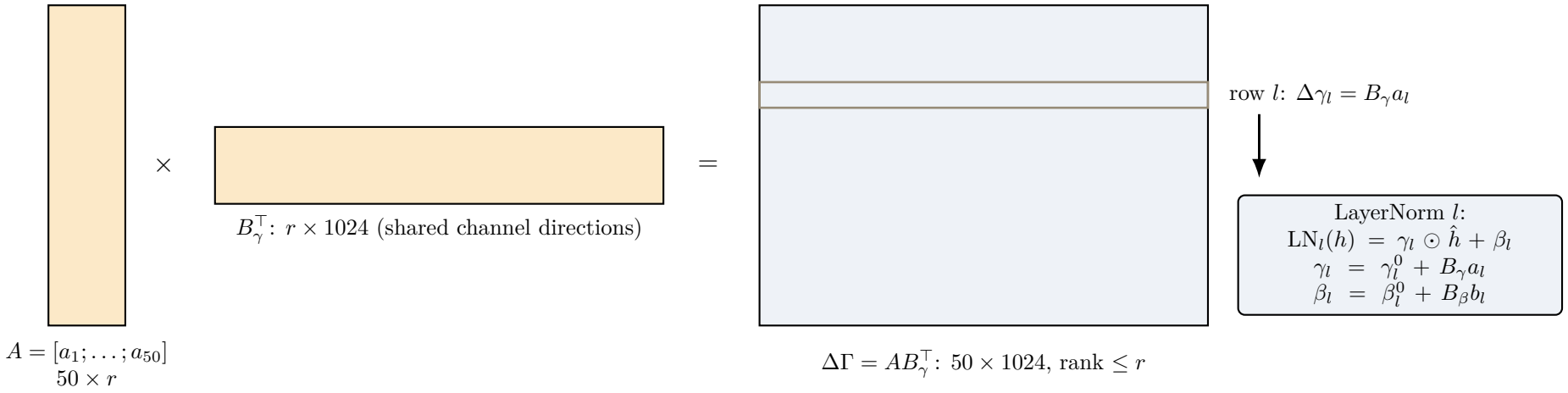
Proposed, not implemented. Orange = trained (17,186 numbers at $r = 8$); blue = frozen; grey = fixed computation or stored buffer.



Replaced from the implemented model: 50 independent LayerNorms (102,400) → SB-LN (17,184 at $r = 8$, 8,592 at $r = 4$); linear head $1024 \to 2$ (2,050) → real-anchored cosine head (2). Fallback if M10b rejects the cosine head: one-direction head $p = \sigma(w^\top z + b)$ (1,025). For fixed features the cosine head's frame AUC does not depend on m_0, τ.

Shared-Basis LayerNorm (SB-LN) and Parameter Budget

All 50 LayerNorm changes are one 50×1024 matrix of rank $\leq r$; the channel directions are shared, each LayerNorm only chooses r coefficients.



Same construction for the shift: $\Delta\mathrm{B} = CB_\beta^\top$, $C = [b_1; \ldots; b_{50}]$. The implemented model is the special case $r = 1024$, $B = I$ (independent LayerNorms). The private offset of generator g lives in coefficient space, $(A + \Delta A_g,\ C + \Delta C_g)$, with the bases shared.

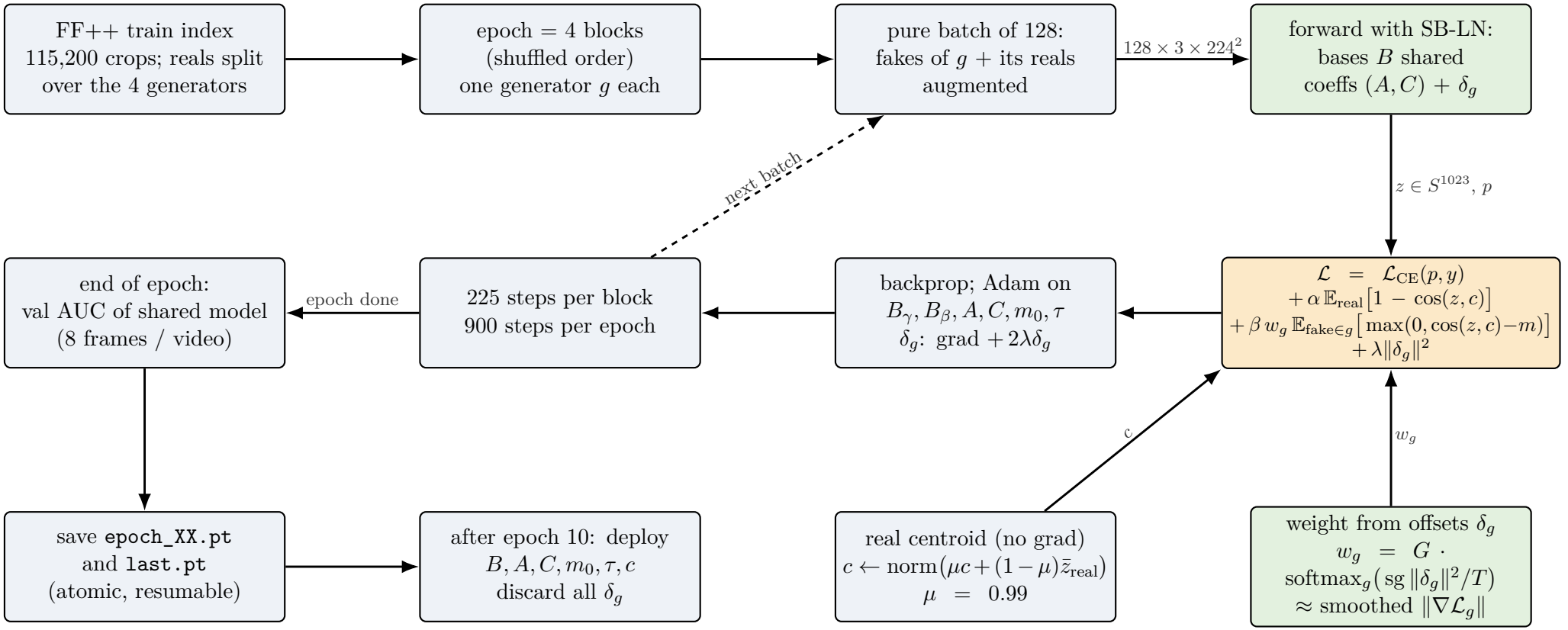
model	adapter	head	trainable (deployed)	frozen	total	trainable %
implemented GBLA (50 LNs + linear head)	102,400	2,050	104,450	303,077,376	303,181,826	0.0345
proposed SB-LN $r = 8$ + cosine head	17,184	2	**17,186**	303,179,776	303,196,962	0.0057
proposed SB-LN $r = 4$ + cosine head	8,592	2	8,594	303,179,776	303,188,370	0.0028
fallback SB-LN $r = 8$ + one-direction head	17,184	1,025	18,209	303,179,776	303,197,985	0.0060

during training	shared	private (4 generators)	stored
implemented GBLA (δ_g in full θ space)	104,450	$4 \times 104{,}450$	522,250
proposed, $r = 8$ (δ_g in coefficient space)	17,186	4×800	**20,386**
proposed, $r = 4$	8,594	4×400	10,194

Reduction of deployed trainable numbers: 6.1× ($r = 8$), 12.2× ($r = 4$); private offsets 130× smaller per generator at $r = 8$. The real centroid c is a stored, non-trained buffer of 1,024 numbers (not counted as trainable). Frozen count of the proposed model includes the 102,400 pretrained γ^0, β^0, which are no longer trained.

Proposed Training Workflow — Coefficient-Space Private Offsets + GBRA Loss

GBRA = Generator-Balanced Real-Anchored loss. The weight of each generator is read from the norm of its private offset, which links the private latent space to the loss.



First-order link: at the optimum of the private objective $\delta_g \approx -\nabla\mathcal{L}_g(\theta_s)/(2\lambda)$, so $\|\delta_g\|^2 \propto \|\nabla\mathcal{L}_g\|^2$: generators the shared model fits worst (M7b: NeuralTextures) get the largest margin weight, at no extra cost. $\alpha, \beta, m, T, \lambda, \mu$ are chosen on FF++ validation in M11; nothing is selected on test data. The cosine head's c and the loss use the same centroid. Planned ablations: $w_g = 1$; loss-based weights (GroupDRO); no private offsets; offsets in full θ space; linear vs. cosine head; $r \in \{4, 8, 16\}$.

Claimed Contributions and Closest Prior Work

Based on our literature search (Nov 2026). Claims marked "pending" depend on M10a / M10b / M11 results.

#	contribution	what is new	closest prior work	evidence
N1	**Shared-Basis LayerNorm (SB-LN)**	One learned r-dim channel basis shared by all 50 LayerNorms; each LN learns only $2r$ coefficients. LN tuning becomes a rank-r matrix factorisation ($102{,}400 \to 2{\cdot}1024r + 100r$).	LN-Tune (AAAI 2024): independent LNs. arXiv 2508.07577: one scalar rescale of LN shifts. ShareLoRA, ASLoRA, RaSA: shared bases for weight matrices, not LN.	M10a (spectrum, truncation)
N2	**Coefficient-space private latent**	Generator-specific offsets live in SB-LN's coefficient space ($100r$, 800 per generator at $r = 8$ vs. 104,450); bases stay shared, so private and shared models differ only in how they mix common directions.	Our GBLA (full-space offsets); M2/M2b random subspaces failed, this subspace is learned.	M10a §4; M11
N3	**Offset-norm generator weighting**	$w_g = G \cdot \mathrm{softmax}(\mathrm{sg}\,\|\delta_g\|^2/T)$: the private offsets act as a free, smoothed per-generator gradient-norm estimate ($\delta_g \approx -\nabla\mathcal{L}_g/2\lambda$) that drives the loss. This is the link between the private latent space and the loss.	GroupDRO (ICLR 2020): weights from group loss. TSRL (2026): RL tutor reweighting. DOSS-Weight (speech). None use private-offset norms.	M7b motivation; M11 ablation vs. GroupDRO
N4	**GBRA loss**	Real-anchored compactness plus generator-balanced margins weighted by N3; the same centroid c is used by the loss and the head, so training and inference match.	Single-Center Loss (CVPR 2021, face forgery): real centre + margin, no generator weighting. OC-Softmax (audio). Centre term credited to them.	M11
N5	**Real-anchored cosine head** (pending)	Classifier with 2 trained numbers $\sigma((m_0 - \cos(z, c))/\tau)$, c a non-trained real centroid; replaces the 2,050-number linear head.	SCL keeps a softmax classifier; OC-Softmax learns the direction. Final check needed before claiming.	M10b
N6	**Parameter efficiency**	17,186 deployed (6.1× fewer) or 8,594 (12.2×); 20,386 stored during training vs. 522,250.	LNCLIP-DF / our GBLA: 104,450.	M11 (accuracy at fixed budget)

Success criteria for M11 (set before training): ≤20k deployed trainable numbers; FaceShifter (unseen) video AUC ≥ 92; DFD ≥ 96; held-out NeuralTextures ≥ 64.4 (the pooled $\lambda = \infty$ level). Carried over from the implemented work: GBLA shared + private training with only the shared model deployed.